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INCH-POUND

MIL-T-38144D(USAF)
15 MARCH 1995
SUPERSEDING
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12 January 1981

MILITARY SPECIFICATION

TRANSMITTER, RATE OF FLOW, FUEL

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements for turbo-jet engine fuel, rate of flow transmitters as specified herein.

1.2 Classification. Transmitters shall be of the following types, as specified (see 6.2):

Type	Flow range (PPH)
TRU-63A	0 to 80,000
TRU-63/A-2	0 to 20,000
TRU-63/A-3	0 to 80,000
TRU-63A/A-3	0 to 80,000
TRU-63/A-4	0 to 100,000

2. APPLICABLE DOCUMENT

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-B-636 Box, Shipping Fiberboard
PPP-B-1672 Boxes, Shipping, Reusable With Cushioning

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: OC-ALC/TICLA, TINKER AFB, OK 73145-5990 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6620

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MILITARY

MIL-P-116	Preservation, Method of
MIL-B-131	Barrier Materials, Watervaporproof, Greaseproof Flexible, Heat-Sealable
MIL-C-5541	Chemical Conversion Coatings On Aluminum and Aluminum Alloys
MIL-T-5624	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-C-26500	Connector, General Purpose, Electrical Miniature, Circular Environment Resisting, Established Reliability
MIL-I-38143	Indicator, Rate of Flow, Fuel
MIL-M-45202	Magnesium Alloy, Anodic Treatment of

STANDARDS

FEDERAL

FED-STD-101	Preservation, Packaging And Packing Materials: Test Procedures

MILITARY

DOD-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Standards and Specifications Order of Precedence for the Selection of
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-810	Environmental Test Methods
MIL-STD-882	Sysytem Safety Program for Systems and Associated Subsystems and Equipment: Requirements for
MIL-STD-889	Dissimilar Metals
MIL-STD-2073	DOD Material Procedures For Development And Application Of Packaging Requirements
MS24265	Connectors, Receptacle, Electric-Single Hole Mount, Miniature
MS24266	Connector, Plug, Electrical-Straight, Miniature
MS27034	Connector, Receptacle, Electrical, Pin Insert, Cylindrical, Miniature

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order desk, Bldg 4D, 700 Robbins Ave, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Transmitters furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.4 and 6.3).

3.2 Selection of standards and specifications. Standards and specifications for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-970.

3.3 Materials

3.3.1 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the transmitter except where magnetic materials are essential.

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3.3.2 Nonferrous materials. Nonferrous materials shall be used for all parts of the transmitter except where ferrous materials are essential.

3.3.3 Metals. Metals shall be of the corrosion-resistant type or treated to resist corrosion due to fuels, salt fog, or atmospheric conditions likely to be met in storage or normal service.

3.3.4 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.3.5 Nonmetallic materials. All nonmetallic materials used for sealing, electrical insulation or other purposes, shall not support the growth of fungus as encountered in tropical areas.

3.3.6 Toxic and corrosive fumes. The transmitter shall be of such material and construction that it will not produce corrosive or deleterious fumes, acid, or alkali under normal use or under the test conditions specified herein.

3.3.7 Protective treatment. When materials are used in the construction of the transmitter that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic or environmental conditions shall be avoided.

3.3.8 Titanium. Where high temperature and other stringent environmental conditions are a factor in the design of the transmitter, the use of titanium will be permitted, with the approval of the procuring activity, where lightweight and high-strength materials are required. Use of lightweight materials and weight-saving designs shall be of major consideration and their use shall be investigated and exploited to the greatest possible extent.

3.4 Design and construction. Unless otherwise specified, the requirements of this specification apply to all types of transmitters specified herein.

3.4.1 Fuel variation. The transmitter shall be designed to operate satisfactorily with fuels having any of the characteristics specified in MIL-T-5624. The transmitter shall be calibrated for rate of flow in pounds per hour and shall measure the fuel flow rate on a gravimetric basis without separate and distinct density correcting means.

3.4.2 Output signal. All types of transmitters shall provide a five-wire synchro type output which shall have electrical characteristics conforming to Clifton Precision Products type TGC-11-E-6 synchro torque or approved equivalent.

3.4.2.1 Output load error. When two indicators are operated from one transmitter output, the resulting load errors shall not exceed 1° of arc more than the errors resulting when only one indicator is used.

3.4.3 Shaft rotation. For an increasing flow rate, the direction of rotation of the transmitter synchro shaft shall be clockwise (cw) as viewed from the shaft end. The total angular travel shall be 160°.

3.4.4 Power. The transmitter shall be designed to operate from 115V 400-Hz, ±5 percent, single-phase ac aircraft power and as otherwise stated in MIL-STD-704. The wave form may be nonsinusoidal with a crest factor of 1.414 ±10 percent and with harmonics with respective amplitudes of less than 5 percent. No damage to the transmitter shall result if a voltage and frequency of 10 percent more or less than the tolerances specified is applied.

3.4.5 Pressures. The TRU-63/A, /A-3, A/A-3 and /A-4 transmitters shall withstand a static fuel pressure of 450 psi at room temperature and at -54°C, and 400 psi at an ambient temperature of 150°C. The TRU-63/A-2 transmitter shall withstand a static fuel pressure of 1,600 psi at room temperature and at -54°C, and 800 psi at an ambient temperature of 177°C.

3.4.6 Electrical receptacles. The transmitter shall be provided with a hermetically sealed, miniature electrical receptacle in accordance with MIL-C-26500 and shall be designed to meet the environmental requirements specified herein.

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3.4.7 Internal wiring. Where practicable, the internal wiring shall be color coded. The receptacle shall be marked as follows:

Pin	Connection
2	Power input 115V 400-Hz to transmitter motor and synchro rotor
1	Power ground side
3	X_2 of transmitter synchro stator
5	Y_2 of transmitter synchro stator
4	Z_2 of transmitter synchro stator
6	Spare
7	Spare

3.4.8 Electrical zero. Identification of the synchro leads and the determination of electrical zero shall be as follows:

a. The rotor leads shall be designated H_2 and C_2, and the C_2 lead shall be grounded

b. One stator lead shall be designated Z_2 and connected to the C_2 lead

c. Rated voltage shall be applied across H_2 and C_2, and the voltage measured across H_2 and one of the undesignated stator leads and across the two undesignated stator leads.

d. The rotor shall be rotated until the voltage measured across H_2 and the one undesignated stator lead is approximately a minimum and the voltage across the two undesignated stator leads is exactly minimum. This position is the electrical zero position corresponding to one-half full scale indication.

e. The rotor shall be rotated in a clockwise direction from electrical zero as viewed from the rotor shaft end, the voltage across one of the undesignated stator leads and Z_2 shall increase, and the voltage across the other undesignated stator lead and Z_2 shall decrease. The stator lead for increasing voltage shall be labeled X_2 and the stator lead for decreasing voltage shall be labeled Y_2

3.4.9 Pressure drop. The transmitter shall be so designed that when inoperative and under any fuel conditions, the pressure drop will not be appreciably more than when the transmitter is operating normally. The tolerances specified in table I shall apply.

TABLE. I Flow rate and pressure drop.

TRU-63/A, /A-2, & /A-3		TRU-63A/A-3		TRU-63/A-4	
Flow Rate (PPH)	Max. Allowable Pressure Drop (PSI)	Flow Rate (PPH)	Max. Allowable Pressure drop (PSI)	Flow Rate (PPH)	Pressure Drop (PSI)
10,000	0.10	10,000	0.13	17,500	0.1
20,000	0.15	20,000	0.20	100,000	2.5
40,000	0.45	40,000	0.55		

3.4.10 System compatibility. The TRU-63/A /A-3, and A/A-3 transmitters shall be completely compatible with EFU-22A/A, A/A-2 and A/A-7 indicators; the TRU-63/A-2 transmitter shall be completely compatible with EFU-22A/A-5 and A/A-6 indicators.

3.4.11 Reliability. The transmitter shall have minimum mean-time-between failure (MTBF) of 1,200 hours demonstrated to a 90 Percent confidence level.

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3.4.12 Longevity. The transmitter shall have an operating life span (equipment longevity as defined in MIL-STD-781) of not less than 2,500 hours before wearout failures occur or the transmitter consistently fails to meet the specified MTBF index. Parts requiring replacement within this period of time and the normal operating period of such parts shall be reported by the contractor to the procuring activity.

3.4.13 Maintainability. The following qualitative requirements shall be considered in the design of the transmitter:

a. Minimization of the complexity of maintenance tasks (for example: Calibration, adjustments, inspection, et cetera), by maximum use of simple design which includes optimum interchangeability and use of standardized equipment or commercial items.

b. Rapid and positive recognition of equipment malfunction or marginal performance.

c. Rapid and positive identification of the replaceable defective part, assembly or component.

d. Minimum numbers and types of tools and test equipment (special and standard) needed to perform maintenance.

e. Optimum accessibility in all components requiring maintenance inspections, removals, or replacements.

3.4.14 Safety. The transmitter shall present no danger, injury, or hazard to operating and service personnel. Provisions of MIL-STD-454, Requirement 1 shall apply.

3.5 Performance. The transmitters shall be capable of meeting the requirements specified herein under the following conditions:

a. Temperature - TRU-63/A, /A-2, /A-3, A/A-3 and /A-4 - operating temperatures ranging from -54° to +121°C and exposure temperatures ranging from -62° to +190°C with fuel ambients through the transmitter ranging from -54° to +71°C (+90°C for the TRU-63/A-4)

b. Vibration from 0.036 double amplitude to ±12 g's

c. Altitudes ranging from sea level to 80,000 feet

d. Shock forces up to 15 g's

e. Salt fog for a period of 50 hours

f. Humidity up to 100 percent

g. High temperature-pressure leakage

(1) TRU-63/A, /A-3, A/A-3 and /A-4 - Temperature of 177°C at 180 psi for 1/2 hour with the fuel temperature at 71°C

(2) TRU-63/A-2 - temperature of 140°C at 800 psi for 1/2 hour with the fuel temperature at 71°C

h. Voltage and frequency variation ranging from 109V 380 Hz to 121V 420 Hz

i. Dielectric strength - application of 500V at 60 Hz between any pin or the electrical receptacle and the transmitter case for 5 seconds.

j. Flame (TRU-63/A-2 and /A-4 only) - withstand a 1,093°C flame for 5 minutes.

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3.5.1 Electromagnetic interference. The transmitter shall meet the electro-magnetic interference and susceptibility requirements of MIL-STD-461 as specified for class A1 equipment.

3.5.2 Friction error. Friction error shall be within the tolerance specified in table II.

3.5.3 Scale error. The transmitter shall meet the scale error tolerances specified in table III.

3.5.4 Power consumption. Power consumption shall not exceed 40 va with nominal power applied to the transmitter.

3.5.5 Response. The time required for the transmitter output signal to come within 1 percent of the final steady state signal shall not exceed 5 seconds under the conditions specified in 4.6.6.

TABLE II. Transmitter accuracy.

TEST	Tolerance (±PPH)	
	63/A, /A-3 A/A-3 & /A-4	63/A-2
Voltage and frequency error	5 percent of point	5 percent of point
Position error	50	25
Friction		
Normal, room, high, and low temp	50	20
Vibration		
Oscillation	100	50
Variation	100	50

3.5.6 Accuracy. The transmitter shall operate within the tolerance specified in tables I, II, and III.

3.5.7 Contaminated fuel. The transmitter shall operate as specified herein with contaminated fuel containing foreign particles of the size specified in table IV.

3.6 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of DOD-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.7 Over-flow. The transmitters shall not be damaged by fuel-flow rates exceeding 20 percent of the rated value provided the pressure rating of the transmitters is not exceeded.

3.8 Flow direction. The direction of flow through the transmitter shall be permanently and legibly marked by an arrow as shown on figures 1 through 4.

3.9 Caps. The fuel-line connections shall be provided with protective shipping caps or plugs to prevent foreign material from entering the transmitter.

3.10 Mounting. The transmitters shall be securely mounted to the airframe or engine by means of the end fittings or transmitter frame clamps as required to withstand the most severe aircraft environment. Mounting during testing shall be equivalent to the aircraft mounting.

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3.11 Case. Outline dimensions of the case shall be in accordance with figures 1 through 4, as applicable.

3.12 System interconnection. System interconnection shall be as shown on figure 5 or 6, as applicable.

3.13 Weight. The weight of the TRU-63/A, /A-2, /A-3 and A/A-3 transmitters shall not exceed 7 pounds; the weight of the TRU-63/A-4 transmitter shall not exceed 7.7 pounds.

3.14 Dimensions. The outline dimensions of the transmitter shall conform to figures 1 through 4, as applicable.

3.15 Finishes and protective coatings

3.15.1 Alloy parts. Where practicable, aluminum or titanium alloy parts shall be covered with an anodic film conforming to MIL-A-8625. Small holes, inserts, and small areas that must be machined after finishing need not be covered with a protective film.

3.15.2 Chemical treatment. Where the primary purpose of the treatment is to afford a suitable paint base, or for castings containing nonaluminum or titanium alloy inserts, a chemical treatment in accordance with MIL-C-5541 may be used in lieu of anodic film for alloys that do not anodize satisfactorily.

3.15.3 Abrasion resistance. Where abrasion resistance is a factor, chemical film in accordance with MIL-C-5541 shall not be used.

3.15.4 Magnesium parts. Magnesium alloy parts shall be covered with an anodic film in accordance with MIL-M-45202, HAE process.

3.16 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.17 Workmanship The transmitter shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, welding, riveting, brazing, plating, painting, machine screw assemblies, and freedom of parts from burrs and sharp edges. Gears shall be of high quality and shall provide a minimum of backlash necessary for smooth operation. Where necessary, antibacklash techniques shall be used.

3.18 Recycled and reclained materials. The use of recycled and recycled and reclaimed materials shall be encouraged to the maximun extent possible without jeopardizing the intended end use of the item .

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TABLE III. Scale error tolerance.

Fuel Flow (PPH)	Tolerances (±PPH) Room Temp			Tolerance (±PPH) High & Low Temp Substitute Fuel				
	TRU-63/A & A-3	TRU-63A/ A-3	TRU-63/ A-4	TRU-63/ A-2	TRU-63/A & /A-3	TRU-63A/ A-3	TRU-63/ A-4	TRU-63/ A-2
0 - 500 *#	250	250	250	120	350	350	350	150
1,000 *#	150	150	100	50	175	175	125	70
2,000 *#	80	80	50	25	100	100	75	40
3,000 *#				30				50
4,000 *#	50	50	50	30	80	80	75	50
6,000 *#	80	80	50	50	110	110	75	70
9,000 *#	100	100	80	80	130	130	120	100
10,000								
12,000 *#	120	120	100	110	150	150	150	150
15,000	120	120	100	140	180	180	150	180
20,000 *#	150	150	120	190	220	220	175	240
25,000	150	150			230	230		
30,000 #	180	150	140		250	240	200	
40,000 *#	360	200	300		440	290	375	
50,000 #	460	250	375		550	340	470	
60,000 *#	560	560	450		650	650	560	
70,000	660	660	525		750	750	655	
80,000 *#	760	760	600		850	850	750	
100,000			750				940	

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~ For acceptance tests on TRU-63/A-4

Yor acceptance tests on TRU-63A/A-3

TABLE IV Contaminated fuel particle size.

Foreign Particle Size (Microns)	Percent of Total Weight
0 - 5	39 ±2
5 - 10	18 ±3
10 - 20	16 ±3
20 - 40	18 ±3
over 40	9 ±3
Through a 200-mesh screen	100 percent

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as speciried herein. except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assur-supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's

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overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or suppliessubmitted to the Government for acceptance comply wit;h all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of tests. The inspection and testing of transmitters shall be classified as follows:

a. Qualification tests

b. Sampling plan (A or C)

c. Individual (acceptance tests)

4.3 Test conditions

4.3.1 Test power. The supply power shall be single-phase ac, 115V ±2 percent, and 400 Hz controlled to less than +1 Hz and of known value.

4.3.2 Test fuel. Unless otherwise specified, the fuel used for calibration during testing shall be JP-4 conforming to MIL-T-5624. A calibration fluid confirming to MIL-C-7024 type II may be used as an alternate. Fuel .shall not flow during testing unless otherwise specified.

4.3.3 Test position. Unless otherwise specified, the transmitter shall be tested with the flow axis horizontal and the connector at the 3 o'clock position when viewed from the inlet end.

4.3.4 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of the tests are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 25°C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowances shall be made for the difference from the specified conditions.

4.3.5 Vibration. Unless otherwise specified, the transmitter .shall be vibrated before a test reading is taken. A vibrator, set at 3,600 Hz, with a maximum amplitude of 0.002 inch and attached to the flow pipe may be used.

4.3.6 Test equipment standards

4.3.6.1 Transmitter output indications. A phase sensitive nullmeter with damping and related null balance equipment shall be standard readout equipment for the fuel-flow transmitter or other readout equipment as approved by the procuring activity. All qualification and sampling plan B flow tests shall be performed on a primary flow stand which measures liquid mass flow rate directly in terms of weight and time. The combined test equipment shall have a total maximum error of 0.4 percent of point.

4.3.7 Tolerances. Test tolerances shall be as specified in tables I, II, and III. When tolerance for fuel temperature or other temperatures are not specified the tolerance shall be ±3°C.

4.4 Qualification testing

4.4.1 Test samples. The qualification test samples shall consist of at least seven transmitters representative of the production equipment. Three of the samples shall be subjected to the sampling plan A and B tests and seven samples (may include the three subjected to the sampling plan A and B test) shall be subjected to the sampling plan C tests. The samples shall be identified with the manufacturer's part number.

4.4.3 Qualification tests. The qualification test shall consist of all the tests specified under 4.6. Unless otherwise specified by the qualifying activity, tests shall be conducted in the order listed.

4.5 Quality conformance tests. Quality conformance tests shall consist of:

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a. Individual tests

b. Sampling tests.

4.5.1 Individual tests. Each transmitter shall be subjected to the following tests as described under 4.6:

a. Examination of product

b. Dielectric strength

c. Scale error

d. Friction error

e. Room temperature leakage

f. Response time

g. Burn-in

4.5.2 Sampling tests

4.5.2.1 Sampling plan A. A minimum of three transmitter shall be selected at random from each 100 or fraction thereof produced and subjected to the following tests as described under 4.6:

a. Individual tests

b. Position error

c. Voltage and frequency variation

d. Power consumption

e. Low temperature

f. High temperature

g. Vibration

h. Transmitter pressure drop.

4.5.2.2 Sampling plan A. Unless otherwise specified (see 6.2), three transmitters shall be selected at random from the first 10 produced on the contract or order and subjected to the following tests as described under 4.6:

a. Sampling plan A tests

b. Low temperature exposure

c. Temperature-altitude

d. High temperature exposure

e. Altitude

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f. Cold temperature leakage

g. High temperature leakage

h. Humidity

i. Salt fog

j. Electromagnetic interference

k. Shock

l. Contaminated fuel

m. Substitute fuel

n. Flame test (TRU-63/A-2 and /A-4 only).

4.5.2.2.1 Test schedule. When sampling plan B tests are not specifically waived on a contract, the tests shall begin as soon as the first 10 items have been produced but before any production deliveries have been made (see 6.2.1).

4.5.2.3 Rejection and retest. When an item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected. The contractor shall explain to the Government representative the cause of failure and the action taken to preclude recurrence. After corrections have been made, all tests shall be repeated unless otherwise specified by the procuring activity.

4.5.2.3.1 Individual tests may continue. For production reason, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or produced later shall not be made until it is determined that all items meet all requirements of the specification.

4.5.2.4 Sampling plan C. Seven transmitters shall be selected at random from the first 14 produced and subjected to the following tests as described under 4.6. Rejection for reliability shall be in accordance with 4.6.25.6.

a. Individual tests

b. Reliability

c. Longevity.

4.5.2.4.1 Reliability test program. Within 90 days after receipt of the contract or purchase order and 30 days prior to the scheduled date of invitation of testing, or at such later date as the contracting officer may authorize, the contractor shall submit a detailed reliability test program to the procuring activity for approval. Testing shall begin only after formal approval of this test program by the procuring activity.

4.5.3 Defects in items already accepted. The investigation of the test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and the method of correcting them.

4.6 Test methods

4.6.1 Examination of product. The transmitter shall be inspected to determine compliance with the requirements specified herein with respect to materials, finish, workmanship and marking. Each unit shall be weighed.

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4.6.2 Dielectric strength. A potential of 500V at a commercial frequency shall be applied between any ungrounded pin and the transmitter case for a minimum of 5 seconds. There shall be no insulation breakdown as a result of this test.

4.6.3 Scale error. Scale error shall be determined at the test points specified in table III for both increasing and decreasing flow rates. The increasing flow rate shall be measured by bringing the flow up to but not exceeding the specified rate. The flow shall be decreased to but not below the specified rate. Up-scale and down-scale readings shall not exceed the tolerances specified in table III.

4.6.4 Friction error. Friction error shall be determined both up-scale and down-scale, at the 4,000- and 9,000-pph flow rates. The difference in readings before and after gently vibrating the transmitter shall not exceed the values specified in table II. This test may be combined with the test specified in 4.6.3.

4.6.5 Leakage

4.6.5.1 Room temperature leakage. The transmitter shall be tested for leakage as follows: A nonshock hydraulic pressure shall be applied to the inlet connection with the outlet connection bled free of air and capped. The applicable pressure specified in 3.4.5 shall then be rapidly applied five times at room temperature. After the last application of pressure and while still at the test pressure and temperature, there shall be no indication of leakage for a period of 5 minutes as checked at constant transmitter fluid volume with a pressure gage readable to 2 psi.

4.6.5.2 Cold temperature leakage. The test of 4.6.5.1 shall be conducted at cold temperature as follows: The transmitter temperature shall then be reduced to 54° ±2°C by submersion in a cold liquid bath, already at this temperature, pressure shall be rapidly applied five times. After the last application of pressure, there shall be no evidence of leakage for a period of 5 minutes as detected by the same means used for the room temperature leakage test.

4.6.5.3 High temperature leakage. The high temperature leakage test shall be conducted as follows: A nonshock pressure shall be applied to the inlet connection of the transmitter with the outlet connection bled free of air and capped. The transmitter shall be heated to the temperature specified in 3.4.5 (150°C or 177°C) and the pressure specified therein, as applicable, shall be applied rapidly five times. After the last application and while still at the test temperature and pressure, there shall be no indication of leakage for 5 minutes as checked by monitoring the pressure within the transmitter at constant fluid volume with a pressure gage readable to 2 psi. No operational damage shall result from this test.

4.6.6 Response time. The time required for the transmitter output signal to come within 1 percent of the final steady-state signal shall not exceed 5 seconds with a step change in flow rate from 500 to full-scale indication.

4.6.7 Burn-in. The transmitter shall be filled with fuel, placed in a chamber with the temperature maintained at 50°C and have 115V 400 Hz power applied for a minimum of 24 hours. The transmitter shall be returned to room temperature for a minimum of 2 hours after which it shall meet the test specified in 4.6.3.. If the burn-in test is conducted other than as specified herein, it shall be subject to the approval of the procuring activity.

4 6.8 Position error. Position error shall be determined at the 4,000-pph flow rate. Readings shall be taken in the normal position and with the transmitter turned 90° in any direction. The reading taken in any position shall not differ from the reading taken in the normal position by more than the value specified in table II.

4.6.9 Voltage and frequency variation. This test shall be conducted at 5, 50, and 90 percent of full-scale flow rates. The reading taken at each of the voltage and frequency conditions specified herein shall not differ from the reading taken at the normal condition in excess of the value specified in table II. The applied voltage and frequency for this test shall be set at the following points:

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Voltage	Frequency (Hz)
115	400
109	380
109	420
121	380
121	420

4.6.10 Power consumption. With 115V 400 Hz applied to the transmitter and fuel flowing at 80 percent ±5 percent of the maximum flow rate, power consumption shall not exceed 40 va.

4.6.11 Low temperature. The transmitter shall be placed in a test chamber and maintained at a temperature of -54° ±2°C for a period of 4 hours. At the end of this period and while still at the specified low temperature, the transmitter shall be subjected to the tests specified in 4.6.3 and 4.6.4. The errors shall not exceed the tolerances specified in tables II and III, respectively. The fuel temperature shall be -54° ±2°C during this test.

4.6.12 High temperature-TRU-63/A, /A-2, /A-3, A/A-3 and /A-4 transmitters. While inoperative, the TRU-63/A, /A-2, /A-3, A/A-3 and /A-4 transmitters shall be subjected to a temperature of either 70°C or 90°C, as applicable, for a period of 4 hours after which they shall be tested for scale error and friction error, as specified in 4.6.3 and 4.6.4, at the specified high temperature. The errors shall not exceed the tolerances specified in tables II and III. The fuel temperature for this test shall be 70° ±2°C for the TRU-63/A, /A-2, /A-3, A/A-3 and 90° ±2°C for the TRU-63/A-4 transmitter.

4.6.13 Vibration. With the fuel flow through the transmitter approximately 9,000 pph, a vibration test shall be conducted in accordance with MIL-STD-810, method 514.2, procedure I, equipment category B1 except the following vibration schedule shall be used in lieu of any test curve specified therein:

0.036 double amplitude	5 to	52 Hz
±5g	52 to	130 Hz
±10g	130 to	185 Hz
±12g	185 t0	2,000 Hz

Pointer oscillation and variation shall not exceed the tolerances specified in table II except that a maximum of two resonances will be permitted with an oscillation or variation tolerance of twice that specified therein if oscillation and variation of this magnitude are not obtained simultaneously. At the completion of this test, the transmitter shall be subjected to the scale error and friction error tests. The errors shall not exceed those specified in tables II and III.

4.6.14 Transmitter pressure drop. With a flow rate of 20,000 pph through the transmitter, the pressure drop shall be obtained by sensing the line pressure at the inlet and outlet ends of the flowmeter and then obtaining the difference between the two readings. The pressure drop shall then be obtained in the same manner with a flow rate of 10,000 and 40,000 pph, as applicable, through the transmitter. The pressure drop shall not exceed the values specified in table

4.6.15 Low temperature exposure. The transmitter shall be placed in a test chamber and the chamber cooled to and maintained at a temperature of -62°C for a period of 72 hours at which time the equipment shall be inspected for deterioration. The temperature of the chamber shall then be raised to -54°C and maintained for an additional 24-hour period. At the end of the 24-hour period, while still at -54°C, and with fuel at -54°C, the transmitter shall be subjected to the scale error and friction error tests. The transmitter shall then be maintained at room temperature for a period of not less than 4 hours after which it shall be subjected to the scale error and friction error tests at room temperature. The transmitter shall be examined and there shall be no deterioration of nor damage to the transmitter as a result of extreme temperature exposure.

4.6.16 High temperature exposure

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4.6.16.1 TRU-63/A, /A-2, /A-3 and A/A-3 transmitters. The TRU-63/A, /A-2, /A-3 and A/A-3 transmitters, while inoperative, shall be subjected to a temperature of 121° ±2°C for a period of 4 hours. The temperature shall then be raised to 190° +2°C for a period of 30 minutes. This cycle shall be repeated six times for a total of 24 hours at 121°C and 3 hours at 190°C. The transmitters shall then be energized and immediately tested for scale error and friction error at 70° ±2°C air ambient and 70° ±2°C fuel. The tolerances shall not exceed those specified in tables II and III. There shall be no evidence of transmitter failure as a result of this test.

4.6.16.2 TRU-63/A-4 transmitter. With 44° ±3°C fuel flowing through the transmitter at approximately 30,000 pph, the transmitter shall be subjected to an ambient temperature of 233° ±3°C for a period of 10 minutes. The ambient temperature shall then be raised to 288° ±3°C for a period of 1.0 minute. This cycle shall be repeated six times for a period of 1.0 hour at 233° ±3°C and 6.0 minutes at 288° ±3°C. The transmitter shall then be energized and immediately tested for scale error and friction error at 149° ±3°C air ambient and 91° ±3°C fuel. The errors shall not exceed the tolerances specified in 3.5.3 and 3.5.2 respectively. There shall be no evidence of transmitter failure as a result of this test.

4.6.17 Altitude. The transmitter shall be filled with fuel, placed in an altitude chamber, and subjected to a pressure equivalent to an altitude of 80,000 feet for a period of 4 hours. During this period, the transmitter shall be energized and tested for room temperature scale error at 4,000 pph. Scale error shall be within the tolerance specified in table III.

4.6.18 Humidity. The transmitter shall be subjected to a humidity test in accordance with MIL-STD-810, method 507.1, procedure 1 except that the relative humidity shall be 100 percent. After completion of the cycling, the transmitter shall be subjected to the tests specified in 4.6.3 and 4.6.4.

4.6.19 Salt fog. The transmitter shall be subjected to a salt fog test in accordance with MIL-STD-810, method 509.1, procedure 1, for a period of 50 hours.

4.6.20 Electromagnetic interference. The transmitter shall be subjected to an electromagnetic interference test in accordance with MIL-STD-462, test methods CE01, CE02, CE03, CE04, CS01, CS02, CS06, RE02, RS01, RS02, and RS03. The upper frequency limit for test method RS03 shall be 150 MHz.

4.6.21 Shock. The shock test shall be in accordance with MIL-STD-810, method 516.2, procedure 1, figure 516.2-2.

4.6.22 Contaminated fuel operation. Test fuel containing dirt to the extent of 80 grams of foreign matter to 1,000 gallons of fuel shall be circulated through the transmitter at approximately 2,000 pph for 1 hour. Scale error and friction error tests shall then be conducted using the contaminated fuel. Operation shall be satisfactory and within the room temperature tolerances. The foreign matter shall consist of not less than 68 percent SiO_2 and shall have a particle-size analysis as specified in table IV.

4.6.23 Substitute fuel. The transmitter shall be subjected to the scale error and friction error tests using JP-5 fuel as specified in MIL-T-5624. These tests should be conducted at room temperature conditions.

4.6.24 Flame (TRU-63/A-2 and /A-4 only). While inoperative, but with a fuel flow of 6,000 pph at 30 psig, the transmitter shall be subjected to a 1,093°C flame for 5 minutes. Any physical damage that would result in leakage sufficient to sustain a flame shall be cause for rejection. When sampling plan B and C tests are conducted, this test shall be run after the longevity test.

4.6.25 Reliability The seven transmitters shall be subjected to a reliability test in accordance with MIL-STD-781, test plan IC, which shall demonstrate a minimum acceptable (lower test, 0_1) MTBF of 1,200 hours to a 90 percent confidence level.

4.6.25.1 Test equipment. The test equipment and fixtures shall include provisions for transmitter operation, temperature cycling, and vibration.

4.6.25.2 Operation. During this test, fuel shall be flowing through the transmitter. The transmitter shall be subjected to flow cycles to produce an indication from O to 80 ±5 percent of the transmitter range and back to 0 at a rate of 5 ±1 cpm during the high temperature portion of the temperature cycle. During the low temperature portion of the cycling fuel shall flow at 10,000

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pph. The cycling operation shall be continued until a total of 1,200 hours of ON time has elapsed or until the reliability test has been completed, whichever comes first.

4.6.25.3 Temperature cycling. Unless otherwise specified by the responsible government agency, the fuel temperature during flow cycling shall be as specified below:

a. Starting at room temperature, decrease fuel temperature to -40°C and maintain for 1 1/2 hours.

b. Raise fuel temperature to 70°C and maintain for 1 1/2 hours.

c. Lower fuel temperature to -40°C and maintain for 1 1/2 hours.

d. Repeat steps b and c until reliability requirement has been met. This schedule provides eight 3-hour operational test cycles for each 24-hour period.

4.6.25.4 Vibration cycling. Throughout this test, the transmitter shall be vibrated for a period of 10 minutes during each hour of on-time at any non resonant frequency between 20 and 60 Hz and at a peak acceleration of 2.2g ±10 percent.

4.6.25.5 Performance check. At each 100-hour interval and at the end of this test, the transmitters shall be subjected to the scale error, friction, and response time test. During each scale error test, if any two test points exceed the allowable tolerances for high and low temperature and substitute fuel specified in tables II and III one failure shall be charged against this test.

4.6.25.6 Failure criteria. For an accept decision, each transmitter shall be operated at least 1,000 hours. If a transmitter fails, entries shall be made on the appropriate data logs and the failed equipment shall be removed from the test and repaired with minimum interruption of the transmitters still under test. The procuring activity shall be immediately notified of the failure and the plan for correcting the deficiency. A detailed analysis of the failure will be submitted as soon as it has been completed. No more than a total of 12 failures will be permitted during this test. Acceptance shall be based on the accept-reject criteria specified in MIL-STD-781, paragraph entitled Accept-Reject Criteria for: Test Plan IC.

4.6.25.7 Reporting. Reporting shall be submitted in accordance with MIL-STD-781. The reliability test log and failure summary and analysis shall be included in the final test report (see 4.4.2 and 4.6.25).

4.6.26 Longevity. Testing on two of the seven reliability test samples shall be continued through a longevity test in accordance with MIL-STD-781, test plan XXVII, until 2,500 hours have accumulated on each of the two transmitters.

4.7 Inspection and test. Test of methods of preservation and packing shall be accomplished in accordance with section 4 of MIL-P-116 to insure compliance with section 5 of this specification.

5. PACKAGING

5.1 Preservation. Preservation shall be level "A" or "C" as specified (see 6.2).

5.1.1 Preservation-packaging. Unless other wise specified by the contracting activity, item shall be packaged in Quantity Unit Pack (QUP) of one each. Each item will be provided a preservation method IA-8 In Accordance With (IAW) MIL-P-116.

5.1.2 Cleaning. Item shall be cleaned IAW MIL-P-116 C-1 process.

5.1.3 Drying. Immediately after cleaning the item shall be dried following any one or combination of the drying procedures listed in MIL-P-116. The drying procedures employed shall not be injurious to the item.

5.1.4 Level A. Item shall be preserved IAW MIL-P-116 and MIL-STD-2073 to provide a method IA-8.

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5.1.5 Level C. Item shall be individually preserved in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity as specified in MIL-P-116 and MIL-STD-2073.

5.2 Packing. Packing shall be level "A", "B", or "C" as specified (see 6.2).

5.2.1 Container-cushioning. Unit container requirements shall conform to MIL-B-131, barrier material, unlesscushioni otherwise specified by the contracting activity. Unit containers shall be large enough to allow for application of sufficient cushioning/wrap material, between the item and unit container. Item cushioning/wrap material shall be of sufficient density and thickness to protect the unit container from punctures. Intermediate/shipping containers shall be large enough to allow for application of sufficient cushioning material, between the unit container and intermediate/shipping container. Cushioning material shall be of sufficient density and thickness to insure shock transmission does not exceed peak values in G's established for the item.

5.2.2 Level A. Fiberboard containers do not meet Level "A" container criteria.

5.2.3 Level B. Item will be preserved as specified in 5.1 and shall be packed in exterior containers conforming to PP-B-1672, weather-resistant unless otherwise specified by the contracting activity. Exterior container shall be uniform shape, size and minimum tare and cube, consistent with the protection required. Closure shall be in accordance with appropriate PP-B-1672 procedures and contracting activity.

5.2.4 Level C. Item will be packed in such a manner that will afford adequate protection against physical demage during direct domestic shipment from the supply source to the first receiving activity. These packs shall conform to MIL-STD-2073.

5.3 Marking. Unit, intermediate, and exterior containers shall be marked IAW special markings required by the contracting activity, and MIL-STD-129.

5.4 Inspection and test. Test of methods of preservation shall be accomplished in accordance with section 4 of MIL-P-116 to insure compliance with section 5 of this specification. Packaging tests shall be conducted IAW rough handling as specified in FED-STD-101.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not manadtory.)

6.1 Intended use. The transmitters covered by this specification are intended for use in measuring and transmitting a signal proportional to the total mass rate of fuel flow to the main engine and afterburner of jet aircraft.

6.2 Acquisition requirements. Acquistion documents must specify the following:

a. Title, number, and date of this specification

b. Type of transmitter required (see 1.2)

c. When sampling plan B tests will not be conducted (see 4.5.2.2)

d. That the items specified in 4.5.3 be returned to the contractor for appropriate corrective action at the contractor's expense

e. Levels of packaging and packing required (see section 5).

f. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

g. Any data for which DD Form 1423 have been supplied.

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6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products Lists is the OC-ALC/TICLA TINKER AFB, OK 73145-5990 and information pertaining to qualification of products may be obtained from that activity.

6.4 Subject term (key word) listing.
Motor
Synchro
115 VAC
12 G

6.5 Ordering data. Requisition, contract and orders should specify the part number of the transmitter which are to be procured, and whether overseas packing is desired.

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify

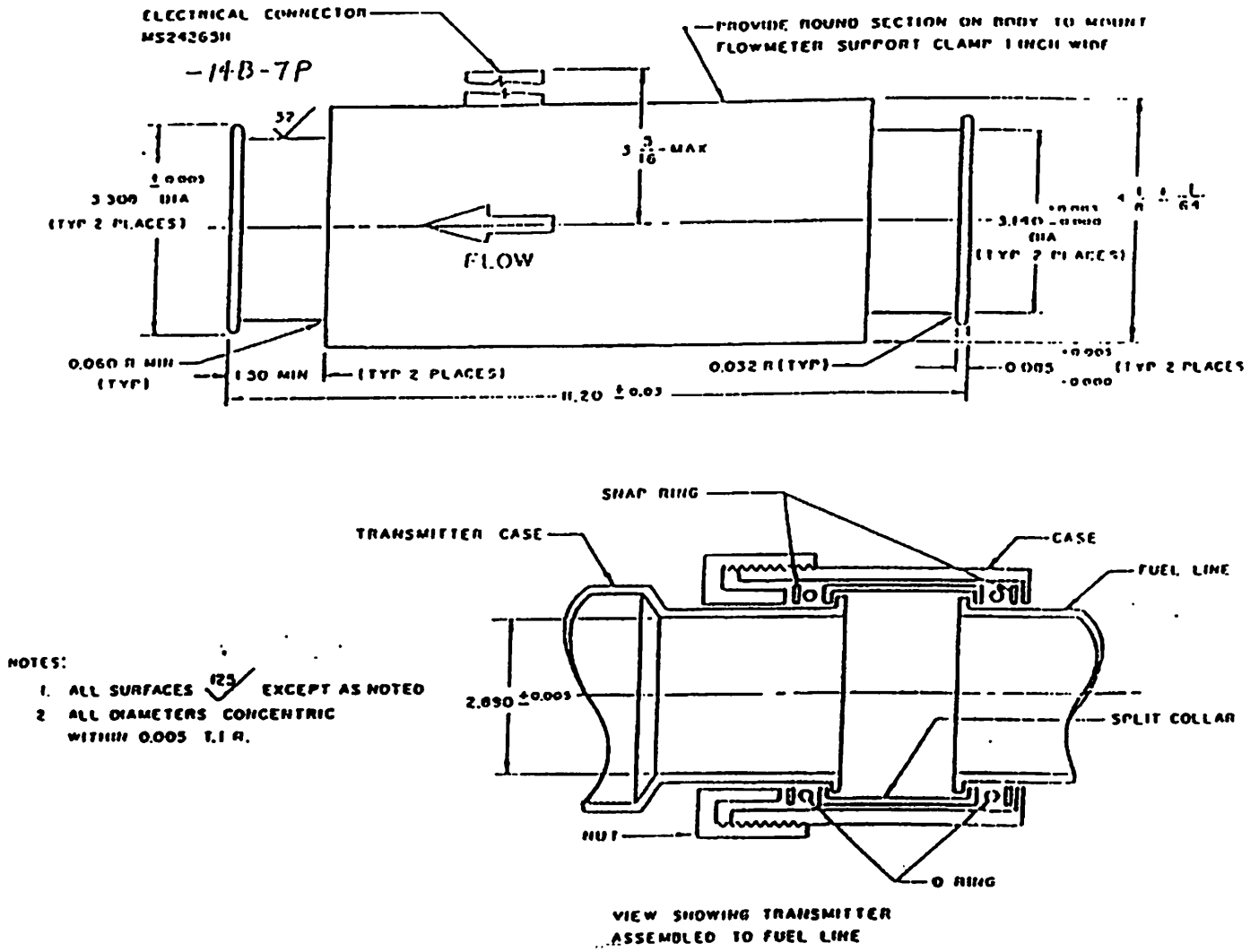
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Preparing activity:
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Review activity:
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NOTES:

1. ALL SURFACES 125 EXCEPT AS NOTED
2. ALL DIAMETERS CONCENTRIC WITHIN 0.005 T.I.R.

FIGURE 1. Outline dimensions, flowmeter TRU-63/A transmitter.

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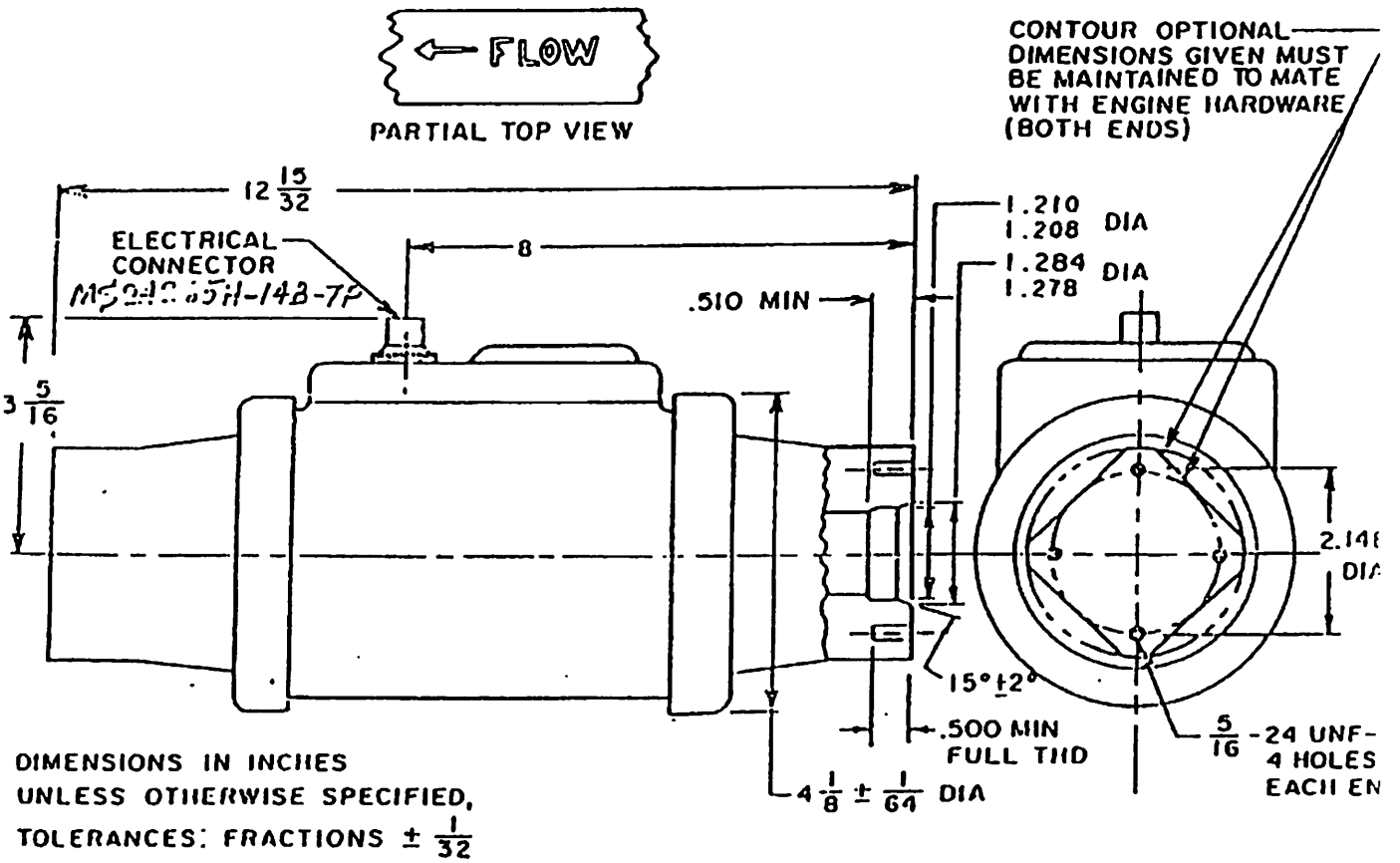
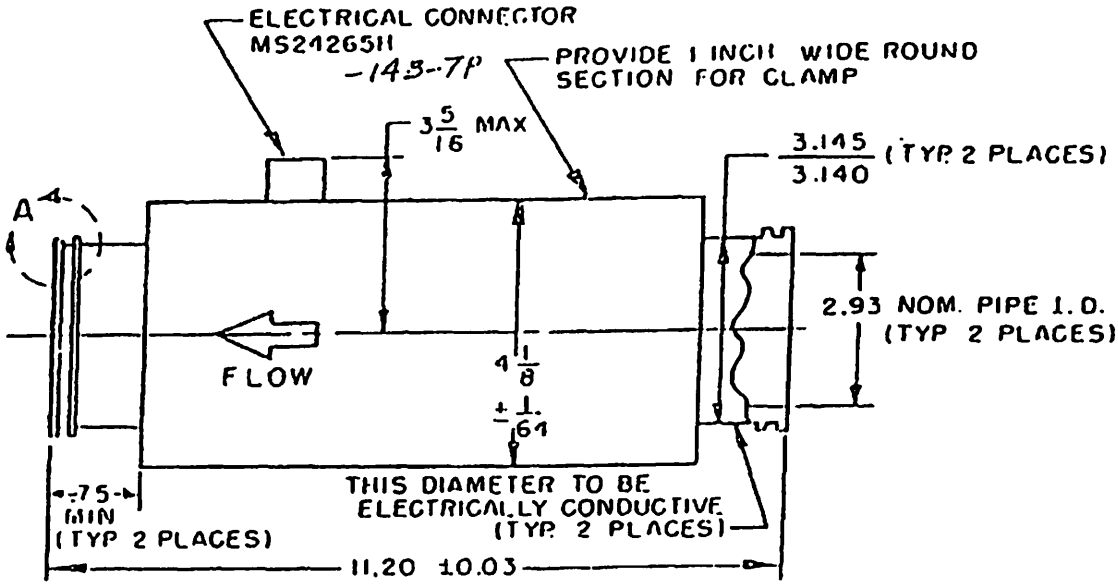
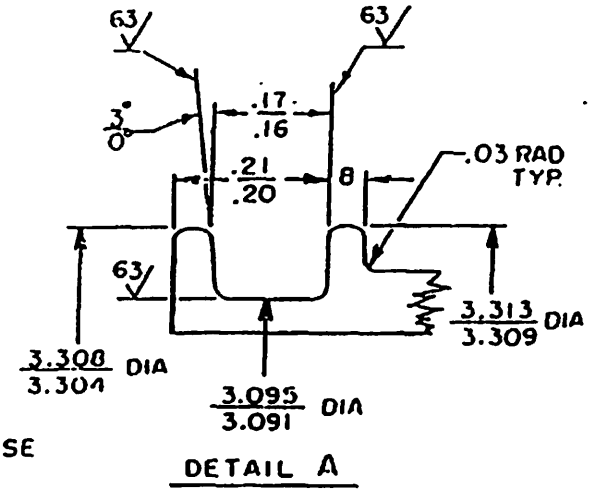


FIGURE 2. Outline dimensions, flowmeter TRU-63/A-2 transmitter.

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1. ALL SURFACES 125 EXCEPT AS NOTED
2. ALL DIAMETERS TO BE CONCENTRIC WITHIN .005 TOTAL.
3. DIMENSION B $\frac{.09}{.08}$ IF CAST, OTHERWISE $\frac{.06}{.05}$
4. DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED.

FIGURE 3. Outline dimensions, TRU-63/A-3 and TRU-63A/A-3 transmitters.

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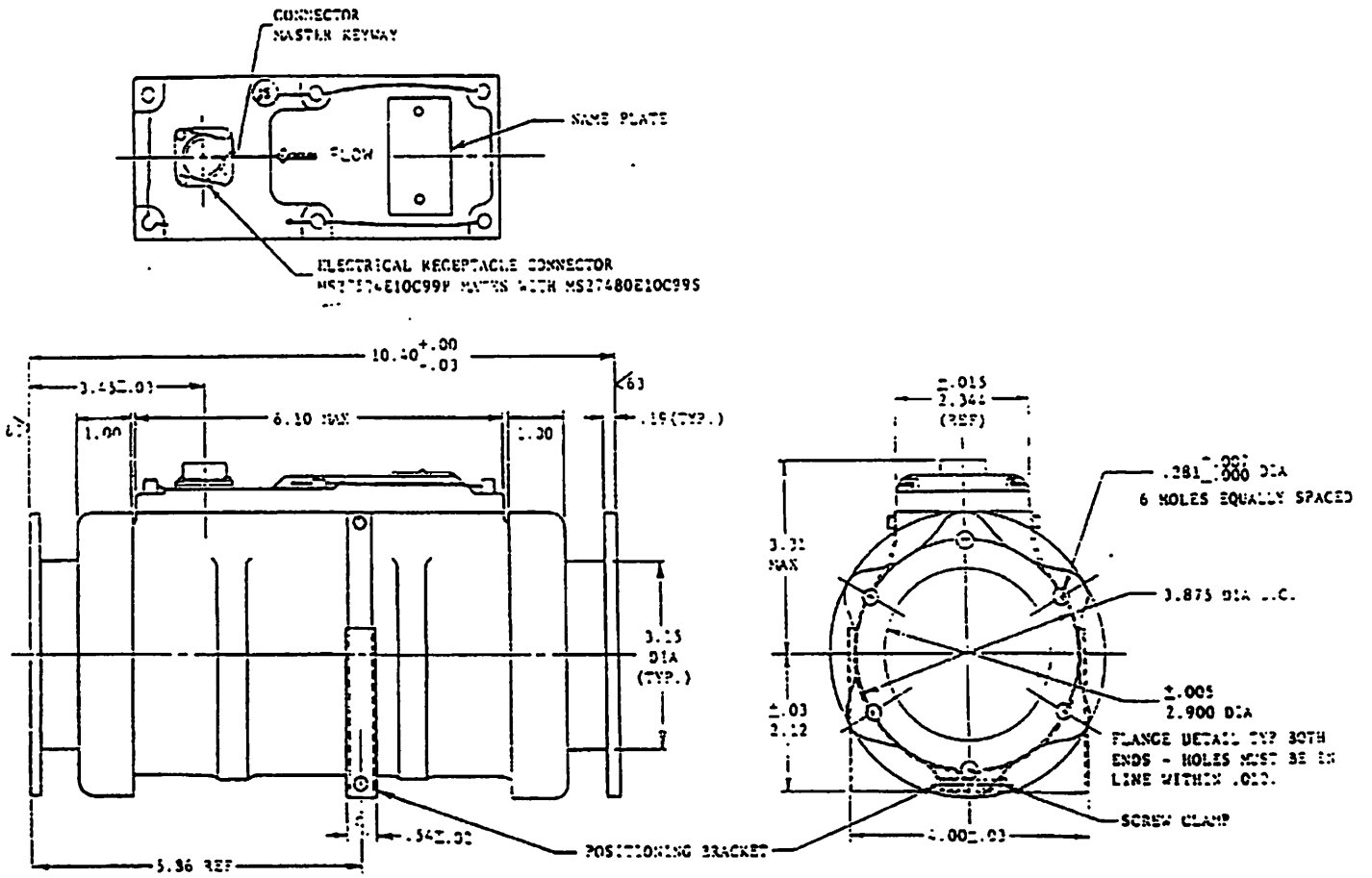
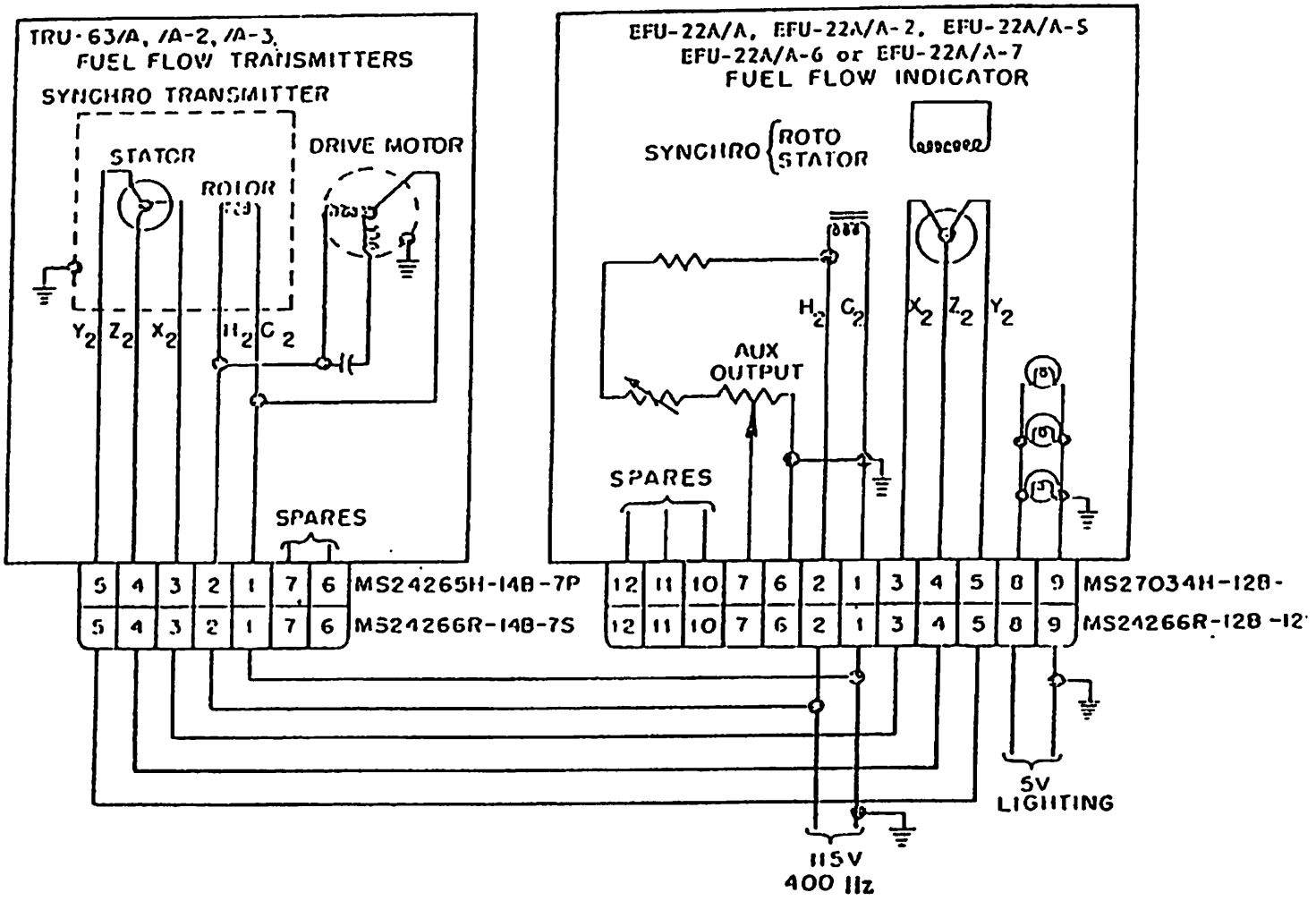


FIGURE 4. Outline dimensions, TRU-63/A-4 transmitter.

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NOTE: THE EFU-22A/A-7 DOES NOT HAVE THE AUXILIARY OUTPUT CIRCUIT AND THERE ARE NO CONNECTIONS TO PINS 6 AND 7.

FIGURE 5. TRU-63/A, /A-2, /A-3, A/A-3 and /A-4 system interconnection diagram.

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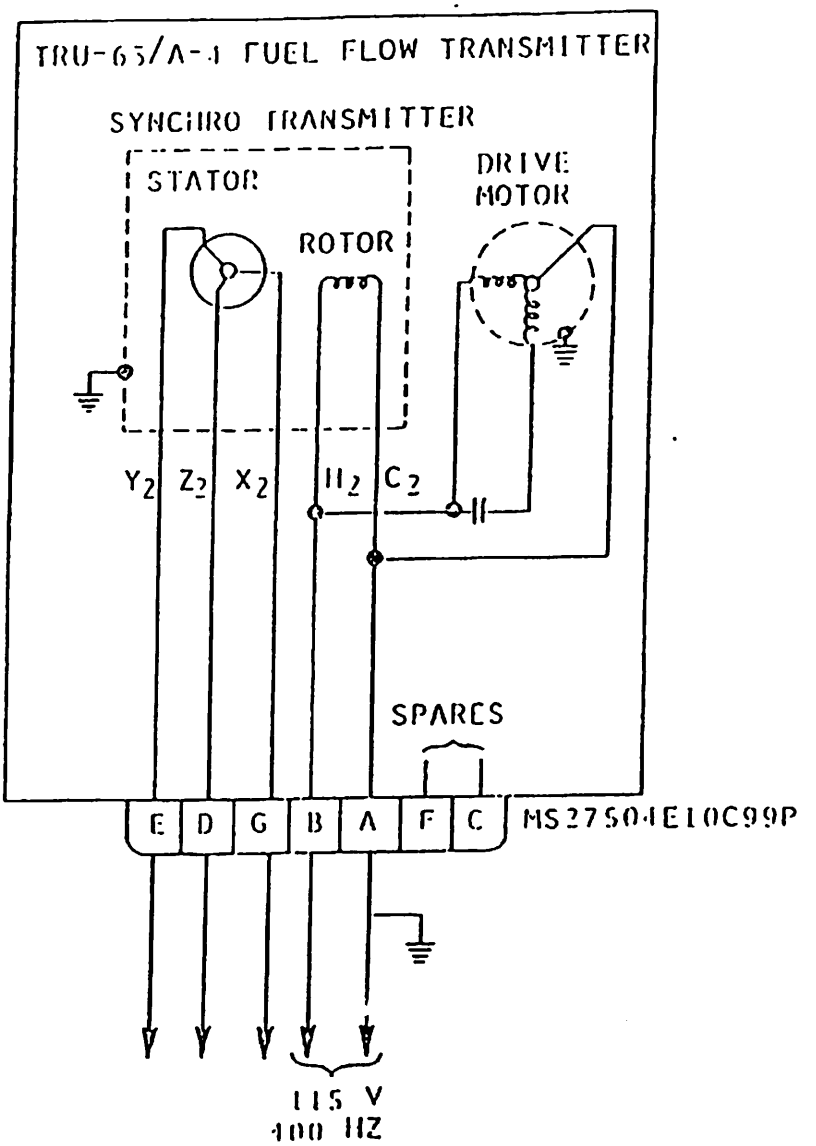


FIGURE 6. TRU-63/A-4 diagram.

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